Inserting a point, subdividing a cell: how the limit is built

For fixed harmonic acceleration $-wx$, $w \geq 0$, arbitrary positive partitions and independently mixed kick–drift orders construct the same motion and force law. The weighted finite estimates in §§3b–3c keep explicit insertion costs, partial-cell conventions and a rational short-window bound; no limiting curve is presumed. The general Lipschitz construction has broader force hypotheses, while these harmonic estimates give sharper constants and the specific source/formalization comparison.

Endpoint elimination, quantum integration and physical recording are different operations. Constant-force elimination and arbitrary-partition Gaussian composition are exact with their stated scalar correction and supplied action coefficient. In $1 + 1$ gauge theory heat-kernel convolution supplies exact surface subdivision; higher-dimensional integration generates boundary interactions whose estimates must persist under iteration. The sufficient consistency norm in §7 does not prove a physical spectral gap or positive-action necessity.

This is a dependency-based exposition of established ingredients and written bounds. Repository model checks are internal; source interpretations and formalization hypotheses are labelled where used. Physical $\hbar$, a time step, lattice spacing and any other regulator remain distinct.

1. What an insertion has to preserve

Three operations occur at a new time r between s and t.

1. A classical calculation adds $q(r)$ and eliminates it by the equations of motion, or by stationary action with fixed endpoints.
2. A quantum propagator integrates over an unobserved intermediate coordinate: $K(t, s) = \int K(t, r)K(r, s)\, dq(r)$.
3. A physical mark introduces an apparatus and a record. Forgetting that record applies the mark's nonselective channel to the body.

Each is a precise composition question. In the third case the nonselective channel must act as the identity on retained observables if one is to recover the unmarked dynamics. Trace preservation alone does not establish that. Thus inserting a variable in a calculation and inserting an actual measurement have separate refinement laws.

For a gauge lattice, multiplying fine links along an old edge defines the old holonomy, while integrating the other variables defines its new probability law. Both the projection of variables and the law must be specified. A choice of fine variables over a coarse configuration also needs a conditional distribution; coarse data alone do not choose that distribution.

2. Newton's one-point insertion, with all constants

Take transverse mass $M > 0$, force F, and a cell of duration $h = u + v$, split after $u > 0$, with $v > 0$. This is the symmetric impulsive construction of the polygon note, a modern constant-force realization of the inscribed-chord comparison. It does not assert that every historical force polygon used this update.

Write a kick and a free drift as

$$K_J(q, p) = (q, p + J), \qquad D_h(q, p) = (q + hp/M, p).$$

The cell map is

$$S_h = K_{Fh/2} D_h K_{Fh/2}, \qquad S_h(q, p) = \left(q + \frac{hp}{M} + \frac{Fh^2}{2M}, p + Fh \right).$$

The input momentum is before the first half-kick and the output is after the last. At a shared vertex the retained momentum is the one between the two adjacent half-kicks; the incoming and outgoing chord momenta depend on the partition.

Proposition 1 (finite classical compatibility). $S_v S_u = S_{u+v}$ for every positive u, v.

Proof. The composed position increment is $up/M + Fu^2/(2M) + v(p+Fu)/M + Fv^2/(2M) = (u+v)p/M + F(u+v)^2/(2M)$; the impulses sum to $F(u+v)$. □

The insertion changes the old endpoint half-kicks as well. Relative to the old two-kick cell, the impulse changes at times $0, u, h$ are

$$-\frac{Fv}{2}, \qquad \frac{Fh}{2}, \qquad -\frac{Fu}{2}.$$

Their total and their first time moment vanish. Consequently the old endpoint position and momentum, with the convention just specified, are preserved. Adding only the middle kick would describe a different force history. The new chord slopes change, while retained positions and the states between half-kicks agree.

The same result can be read directly from a discrete action. Put

$$L_h(x, y) = \frac{M(y-x)^2}{2h} + \frac{Fh}{2}(x+y).$$

Proposition 2 (eliminating the new point). For fixed endpoints x, y, the unique minimizer of $L_u(x, z) + L_v(z, y)$ is

$$z_* = \frac{vx + uy}{h} - \frac{Fuv}{2M},$$

and

$$\min_z \{L_u(x, z) + L_v(z, y)\} = L_h(x, y) - \frac{F^2 uvh}{8M}. \tag{1}$$

Proof. With $\bar{z} = (vx + uy)/h$ and $w = z - \bar{z}$, completing the square gives

$$L_u(x, z) + L_v(z, y) - L_h(x, y) = \frac{Mh}{2uv} w^2 + \frac{Fh}{2} w.$$

The minimum is at $w = -Fuv/(2M)$ and equals the displayed constant. □

That constant is independent of the endpoints, so its derivatives do not change the classical endpoint momenta. It does change the action. Define

$$\ell_h(x, y) = L_h(x, y) - \frac{F^2 h^3}{24M}.$$

Since $h^3 - u^3 - v^3 = 3uvh$, equation (1) becomes the exact law

$$\min_z \{\ell_u(x, z) + \ell_v(z, y)\} = \ell_h(x, y). \tag{2}$$

This constructs the corrected endpoint action from finite insertion compatibility. Two different orders of inserting several points agree: the correction removed is always $F^2(h^3 - \sum_i h_i^3)/(24M)$. Equivalently, the two-step insertion correction satisfies the associativity identity $d(u, v) + d(u + v, w) = d(v, w) + d(u, v + w)$.

Among continuous corrections depending only on the cell duration, the cubic correction is determined up to $E_{\mathrm{ref}} h$: subtracting two solutions of the insertion equation gives the additive Cauchy equation. This remaining freedom changes the zero of energy, leaving spectral differences unchanged. Thus insertion consistency fixes a nontrivial part of the action while retaining the expected energy-normalization freedom.

For the user's example, if $t_{2.6}$ lies at $3/5$ of the old cell, then $u = 3h/5$, $v = 2h/5$. The insertion removes $3F^2h^3/(100M)$, or $18/25$ of that cell's original cubic correction; $7/25$ remains. These are exact fractions.

The geometric interpretation stays with Galileo's inertial line and parabola. At horizontal speed V, the triangle between the old chord and the two new chords has area $VFuvh/(4M)$ for $F, V > 0$; multiplying by $F/(2V)$ gives (1)'s action correction. It is the chord-segment diagnostic of that comparison, not a Kepler swept sector.

3. Quantum composition, and the arbitrary-partition limit

Supply canonical quantum kinematics $[q, p] = i\hbar$, $\hbar > 0$, and use the unitary cell operator

$$Q_h = e^{iFhq/(2\hbar)} e^{-ihp^2/(2M\hbar)} e^{iFhq/(2\hbar)}.$$

Its coordinate kernel has phase $L_h/\hbar$ and free normalization $(M/(2\pi i\hbar h))^{1/2}$, with the usual positive-time branch. Completing the same square as in Proposition 2 in the oscillatory Gaussian integral gives

$$Q_v Q_u = e^{-iF^2uvh/(8M\hbar)} Q_h.$$

The Gaussian normalization composes too: the intermediate integral contributes $(2\pi i\hbar uv/(Mh))^{1/2}$. Hence

$$U_h = e^{-iF^2h^3/(24M\hbar)} Q_h, \qquad U_v U_u = U_{u+v}. \tag{3}$$

This is the constant-force propagator, with generator $p^2/(2M) - Fq$. The scalar correction is measurable as a relative phase when histories are coherently compared. It cancels from the body-only channel $\rho \mapsto Q_h \rho Q_h^\dagger$.

For $F \neq 0$ this mechanical Hamiltonian on the line has spectrum $\mathbb{R}$: unitary translations of q shift it by arbitrary real constants. The present limit is a unitary evolution over a finite experiment. Its action-cost question concerns records of that experiment; the field-theory vacuum spectral gap is a further kind of statement.

The following finite-to-limit statement also covers situations with an error instead of an exact scalar correction. It is an elementary contractive form of a noncommutative sewing argument (Feyel–de La Pradelle–Mokobodzki 2007, abstract; the stronger telescoping hypothesis used here is stated and proved below).

Theorem 3 (summable insertion defects). Let $Q(t, s)$ be bounded operators on a fixed Banach space, $s < t$ in a finite interval, with $\|Q(t, s)\| \leq 1$. Suppose for every $s < r < t$,

$$\|Q(t, s) - Q(t, r)Q(r, s)\| \leq C\big((t - s)^p - (r - s)^p - (t - r)^p\big), \qquad p > 1. \tag{4}$$

For a finite partition π of $[s, t]$, let Q_π be the chronological product. There exists a unique limit $U(t, s)$ as the mesh tends to zero, independent of nesting, unequal cell lengths or insertion order, and

$$\|Q_\pi - U(t, s)\| \leq C \sum_{I \in \pi} |I|^p \leq C(t - s)|\pi|^{p-1}. \tag{5}$$

The limit is contractive and satisfies $U(t, r)U(r, s) = U(t, s)$. If $Q(t, s)$ tends strongly to the identity as $t \downarrow s$, so does $U(t, s)$.

Proof. One insertion changes the full product by at most (4), since all operators before and after that cell are contractions. Successive insertions telescope the potential $\sum_I |I|^p$, so for a refinement π',

$$\|Q_\pi - Q_{\pi'}\| \le C\left(\sum_{I\in\pi}|I|^p - \sum_{J\in\pi'}|J|^p\right).$$

Two arbitrary partitions have a common refinement, their union. Their products differ by at most $C(t-s)$ times the sum of their meshes to the power $p-1$. Completeness gives the limit; refining a fixed π proves (5). Joining partitions on $[s,r]$ and $[r,t]$ proves composition. The one-cell bound $\|U(t,s)-Q(t,s)\| \le C(t-s)^p$ proves the final assertion. □

For constant force, $|e^{ix}-1| \le |x|$ gives (4) with $p=3$ and $C = F^2/(24M\hbar)$. Thus, for duration T,

$$\|Q_\pi - U_T\| \le \frac{F^2T}{24M\hbar}|\pi|^2. \tag{6}$$

This is a constructed arbitrary-partition limit. The analogous estimate for a general nonlinear force needs a suitable domain and stability norm; unbounded commutators need not satisfy (4) in operator norm. The constant-force proof does not silently supply that extension.

Neither (2) nor (3) determines a universal action unit. The classical composition is exact, and the quantum composition works at every supplied $\hbar > 0$. For the Newton necessity goal, the new question is which independently justified physical record or optical law selects the quantum composition structure and a common positive normalization. Refinement of unobserved intermediate variables already works at fixed $\hbar$; an action floor concerns the physical records, not a failure of (6).

3b. Harmonic polygons: a finite refinement bound from a different norm

The sister repository's Proposition I realization graph and HarmonicStability (local full-read at commit `381d5e9`) separate finite stability from convergence of force-generated polygons. The equal-cell quadratic invariant is proved there; a norm inequality in the represented-fraction arithmetic is a pending formalization step. The following written argument uses a weighted sum of absolute values instead. It has not been formalized in that repository and imports no trajectory or ODE existence theorem.

Use its **end-kick** convention, distinct from §2's half-kicks. In one coordinate, acceleration is $a(x) = -wx$, with $w \ge 0$, and a cell of duration h first drifts and then applies the force at the new position:

$$A_h\begin{pmatrix}x\\v\end{pmatrix} = \begin{pmatrix}1 & h\\ -wh & 1-wh^2\end{pmatrix}\begin{pmatrix}x\\v\end{pmatrix}. \tag{6a}$$

For several coordinates apply the same map to each coordinate pair. Choose $\rho > 0$ with $w \le \rho^2$ and define $\|z\|_\rho = \rho\sum_i |x_i| + \sum_i |v_i|$. Let $A_\pi = A_{h_n}\cdots A_{h_1}$ for a finite positive partition of $[0,\tau]$, and assume $\rho|\pi| \le 1$.

Proposition 3b (finite harmonic refinement control). Put $B_\tau = e^{\rho\tau}$. Every such partition satisfies $\|A_\pi\|_\rho \le B_\tau$. If π' refines π, then

$$\|(A_{\pi'} - A_\pi)z\|_\rho \le wB_\tau\left(\sum_{I\in\pi}|I|^2 - \sum_{J\in\pi'}|J|^2\right)\|z\|_\rho. \tag{6b}$$

For any two admissible partitions of the same interval,

$$\|(A_\pi - A_\sigma)z\|_\rho \le w\tau B_\tau(|\pi| + |\sigma|)\|z\|_\rho. \tag{6c}$$

If $\rho\tau < 1$, each occurrence of B_τ in these bounds may instead be replaced by the rational expression $(1-\rho\tau)^{-1}$. Thus on that window the proof uses only finite arithmetic, order and absolute-value inequalities when the data are rational.

Proof. In scaled coordinates $(\rho x, v)$ the absolute column sums of A_h are $1 + wh/\rho$ and $\rho h + |1 - wh^2|$. Since $wh^2 \le 1$, both are at most $1 + \rho h$. The triangle inequality therefore proves $\|A_h\|_\rho \le 1 + \rho h$. Multiplication gives $\|A_\pi\|_\rho \le \prod_j (1 + \rho h_j) \le e^{\rho\tau}$. For $\rho\tau < 1$, expand the finite product: its degree-r elementary symmetric coefficient is at most $(\rho\tau)^r$. The finite geometric sum is at most $(1 - \rho\tau)^{-1}$. This alternative bound applies to every subproduct too.

For a split $h + k$, direct multiplication gives

$$\begin{aligned} \Delta x &= -whk(x + hv), \\ \Delta v &= whk\, v + w^2 hk^2(x + hv), \qquad \Delta = (A_k A_h - A_{h+k})z. \end{aligned} \tag{6d}$$

The absolute column sums of the scaled difference matrix are

$$whk(1 + wk/\rho), \qquad whk(1 + \rho h + whk).$$

They are at most $2whk$: use $w \le \rho^2$ and $\rho h + \rho k \le 1$, which also gives $\rho h + whk \le \rho h + \rho^2 hk \le 1$. Thus $\|A_k A_h - A_{h+k}\|_\rho \le 2whk = w[(h+k)^2 - h^2 - k^2]$. Multiplying by all the earlier and later cell maps costs at most B_τ; their combined duration is at most τ. Successive insertions telescope the sum of squared cell lengths, proving (6b). For arbitrary π, σ use their finite union as a common refinement and $\sum h_j^2 \le \tau|\pi|$, proving (6c). □

The estimate also controls the actual polygon between vertices. At a time t inside a cell, let u be its elapsed duration and let π_t consist of the completed cells followed by u (omit a zero cell). The comparison state $z_\pi^\dagger(t) = A_{\pi_t} z_0$ has exactly the polygon position; its velocity includes a partial kick that the actual polygon has not yet received. For the actual state $z_\pi(t)$, take the outgoing velocity at a completed vertex. Then

$$\|z_\pi^\dagger(t) - z_\pi(t)\|_\rho = wu \sum_i |x_{\pi,i}(t)| \le \frac{w}{\rho} |\pi| B_\tau \|z_0\|_\rho. \tag{6e}$$

Apply (6c) to the two clipped partitions of $[0, t]$, then (6e) twice:

$$\sup_{0 \le t \le \tau} \|z_\pi(t) - z_\sigma(t)\|_\rho \le wB_\tau(\tau + \rho^{-1})(|\pi| + |\sigma|)\|z_0\|_\rho. \tag{6f}$$

This is a finite uniform Cauchy estimate, including unequal schedules and retained velocity. Section 3c completes the state space and identifies the force law. The historical geometric enclosure remains a separate hypothesis. No curve was supplied to prove (6b)–(6f). The norm change supplies a concrete finite input to the graph's P3 obligation, rather than deducing P3 from an area invariant.

There is also a finite input to force identification. Define the trapezoidal position sum, with no integral presumed,

$$T_\pi(t) = \sum_{j \text{ complete}} \frac{h_j}{2}(x_{j-1} + x_j) + \frac{u}{2}(x_{\text{last}} + x_\pi(t)).$$

The same end-kick construction gives the exact identity

$$v_\pi^\dagger(t) - v_0 + wT_\pi(t) = -\frac{w}{2}\left[\sum_{j \text{ complete}} h_j^2 v_{j-1} + u^2 v_{\text{last}}\right]. \tag{6g}$$

Indeed $x_j - x_{j-1} = h_j v_{j-1}$ and the kick is $-wh_j x_j$; subtracting a trapezoid from $h_j x_j$ leaves $h_j^2 v_{j-1}/2$. The partial kick supplies the identical calculation with u. Each departure velocity has absolute sum at most $B_\tau \|z_0\|_\rho$, and $\sum h_j^2 + u^2 \le t|\pi|$. For the actual velocity add the missing partial kick $wux_\pi(t)$ from (6e), obtaining

$$\|v_\pi(t) - v_0 + wT_\pi(t)\|_1 \le wB_\tau(t/2 + \rho^{-1})|\pi|\|z_0\|_\rho. \tag{6h}$$

The position balance $x_\pi(t) - x_0 = \sum_{j\ \mathrm{complete}} h_j v_{j-1} + u v_{\mathrm{last}}$ is exact. At $t = 0$ all sums are empty; at a completed vertex $u = 0$, so the identities retain the declared outgoing velocity. Thus the candidate curve family is paired with a vanishing finite force-balance residual, rather than only a stability estimate. Section 3c passes these sums to the continuous force law; the printed geometric enclosure remains separate.

Printed geometry and formalization scope. The printed-stage edges from Lemma III Cor. 4 to Proposition I record a textual citation. They do not supply convergence of the generated polygon family: the lemma's given-curve approximation and the graph's P3 realization have different inputs. The graph's attached sector_ratio_reconstruction (local full-read) explicitly assumes lower and upper limits and a geometric enclosure. Having that formal reference does not discharge those hypotheses. This check concerns graph metadata and code; the printed passages retain their existing repository attribution. The bounds retain the single B_τ factor and the partial-cell sign in (6g). The finite estimate supplies P3's Cauchy input and (6h) supplies force-consistency data; neither silently closes the graph's limiting geometric or force-identification nodes. This is an internal mathematical check, separate from Lean compilation.

A distinct retained-observable construction is to obtain surviving quantities by repeated block renormalization, without presupposing a limiting physical curve. In this fixed harmonic model the Cauchy bound forces a limit in the usual real-state completion. It does not establish that the full body–apparatus state has the same completion when refinement adds bath and memory variables. For that construction, specify the retained observables and their blocking maps; §7's consistency estimate is a distinct route to their limiting laws. A finite area identity alone establishes neither kind of limit.

Here w has units of inverse time squared and ρ of inverse time; the norm has velocity units, and every coefficient multiplying it in (6b)–(6f) is dimensionless. At $w = 0$ the insertion defect and all partition discrepancies vanish exactly. For fixed τ, w, ρ the bounds tend to zero with both meshes, independently of their nesting. Mass enters only through w if a spring force is written $-mwx$. No action normalization appears. To use this mechanical bound for a record, its discrepancy must still vanish after whitening by the complete conditional record covariance, as tested in the recording note, §6.3.

3c. All refinement sequences and two microscopic kick orders

The user's proposed criterion is independence from **every admitted scaling sequence**, including unequal and nonnested partitions. The local dynamics and physical initial data must be specified throughout. One repeated dyadic sequence alone would not establish that criterion. Here the finite estimate also permits a change of kick order in every cell, and constructs the harmonic curve and its force law.

Retain $w \geq 0$, $\rho > 0$, $w \leq \rho^2$ and the norm of §3b. Let $D_h(x, v) = (x + hv, v)$ and $K_h(x, v) = (x, v - whx)$. The end-kick and start-kick cell maps are respectively

$$A_h = K_h D_h = \begin{pmatrix} 1 & h \\ -wh & 1 - wh^2 \end{pmatrix}, \qquad C_h = D_h K_h = \begin{pmatrix} 1 - wh^2 & h \\ -wh & 1 \end{pmatrix}. \tag{6i}$$

For each cell choose either map, independently of the choices in other cells. Write T_π for their chronological product. Require $\rho|\pi| \leq 1$, which every vanishing-mesh sequence eventually satisfies. Initial position and velocity z_0 and w stay fixed.

Proposition 3c (harmonic scheme universality). With $B_\tau = e^{\rho\tau}$,

$$\begin{aligned} \|T_\pi\|_\rho &\leq B_\tau, \\ \|(T_\pi - A_\pi) z_0\|_\rho &\leq w B_\tau \sum_i h_i^2 \|z_0\|_\rho, \\ \|(T_\pi - T_\sigma) z_0\|_\rho &\leq 2w\tau B_\tau (|\pi| + |\sigma|) \|z_0\|_\rho. \end{aligned} \tag{6j}$$

No nesting or common kick choices are required. Linearly interpolate both vertex state components to obtain a continuous displayed path $\overline{z}_\pi$. Then

$$\sup_{t\le\tau}\|\overline{z}_\pi(t)-\overline{z}_\sigma(t)\|_\rho \le 2wB_\tau(\tau+\rho^{-1})(|\pi|+|\sigma|)\|z_0\|_\rho. \tag{6k}$$

Proof. In scaled coordinates the absolute column sums of C_h are $|1-wh^2|+wh/\rho$ and $1+\rho h$. Under the stated mesh bound they are at most $1+\rho h$, as are those of A_h. Every mixed product consequently has norm at most B_τ. Moreover

$$A_h - C_h = wh^2\,\mathrm{diag}(1,-1), \qquad \|A_h-C_h\|_\rho = wh^2.$$

Replace one factor at a time. Its prefix and suffix have disjoint durations whose sum is at most τ, so their combined amplification is at most one B_τ. This proves the second line of (6j). Combine it for both partitions with (6c) to prove the third.

Interpolation between corresponding vertices preserves the same-partition mixed-versus-end bound. The displayed velocity of an end-kick polygon differs from its actual velocity by at most the cell kick, hence by $(w/\rho)B_\tau|\pi|\|z_0\|_\rho$. Use these two interpolation corrections and (6f), then the two mixed-versus-end comparisons, to obtain (6k). □

The vertex state is the output of the preceding cell and the input to the next. A next-cell start kick occurs immediately after that state. At an end-kick/start-kick interface there are two impulses; the vertex state is between them. Each adjacent velocity differs from the displayed vertex velocity by at most one cell-kick bound. At $t=0$, the displayed initial velocity is v_0; a first start kick changes the outgoing velocity by a quantity vanishing with the mesh. On either kind of cell the actual position equals its displayed linear interpolation, and the actual velocity differs by at most $(w/\rho)B_\tau|\pi|\|z_0\|_\rho$.

Completeness of real continuous path space constructs a limit from (6k), initially on any one vanishing-mesh sequence. The same bound then gives that identical limit for every admitted partition sequence and every sequence of kick choices. Actual velocity representatives converge uniformly too. No limiting curve was used in these estimates.

To identify its force law, first choose an all-end-kick sequence. Its finite position balance and (6h) pass through uniform convergence to

$$x(t)=x_0+\int_0^t v(s)\,ds, \qquad v(t)=v_0-w\int_0^t x(s)\,ds. \tag{6l}$$

The trapezoidal sum in (6h) is exactly the integral of the finite polygon position. The position balance is its finite drift integral. Uniform convergence passes these ordinary integrals to the limit. Thus $x'=v$ and $v'=-wx$. Every mixed construction has this same limit by (6k). This proves trajectory existence for the specified harmonic model; the printed Newton enclosure and the sister Lean formalization of P3 remain separate. At $w=0$, every discrepancy vanishes exactly. The estimates have the units and norm of §3b.

The RG comparison concerns universality of retained descriptions. The finite block map is the full product of its daughter matrices; regrouping that product associates exactly. Replacing it by one bare cell map has a defect, which the estimates control. Thus the curve is obtained from the whole compatible refinement family. Its existence does not require exact equality of the two finite conventions.

The source connection is explicit but has additional hypotheses. In 't Hooft's automaton book, §§5.4 and 20.8 [@tHooft2015CAI] (passages read), he postulates an ontological basis and invokes RG to connect microscopic automaton Hamiltonians to large-scale field theory. Our inference is the universality question: which structures survive a change of finite description? His deterministic ontology is an additional premise. His own publication list (author metadata read) dates the renormalization papers to 1971–73 and deterministic-QM papers to 1988–90; an immediate causal transition is not established by those dates.

Scope. The single B_τ amplification, interpolation and endpoint convention, completion and force-law passage were checked. The 't Hooft attribution is a separate source passage comparison. The result concerns fixed harmonic dynamics, not arbitrary microscopic laws or an action floor.

3d. Scale invariance, cancelled masses and the action that remains

The user's scale comparison adds an exact obstruction to identifying an action unit from curve compatibility alone. An overall action normalization can disappear from every classical trajectory equation and remain in canonical momentum, energy and recorded phase.

Proposition 3d (classical normalization under blocking). For a discrete action S_π and any dimensionless constant $c_* > 0$, stationary configurations of $c_* S_\pi$ are unchanged, and

$$\inf_z c_* S_\pi(x, z, y) = c_* \inf_z S_\pi(x, z, y). \tag{6m}$$

The minimizing configurations, when attained, are the same. This follows directly by multiplying the first variation and the order inequalities by c_*. Consequently any classical insertion or blocking law built from these configurations commutes with this rescaling. Refinement independence of their curve fixes no overall action normalization.

For a free body, $S = m \int |\dot{q}|^2 dt/2$, $m > 0$; fixed initial position and **velocity** give a trajectory independent of m. Momentum $p = m\dot{q}$ and action retain it. For uniform acceleration, put $F = ma$ in §2's corrected cell action:

$$\ell_h(x, y) = m \left[\frac{(y-x)^2}{2h} + \frac{ah}{2}(x+y) - \frac{a^2 h^3}{24} \right]. \tag{6n}$$

The curve and point elimination are independent of the overall m; the cubic action correction is $ma^2h^3/24$. Mass cancellation in the motion therefore leaves the phase-space action question intact.

For a test body in a prescribed external gravitational potential, assume equality of its passive gravitational and inertial masses. On a collision-free interval,

$$S = m \int \left[\frac{|\dot{q}|^2}{2} + \frac{GM}{|q|} \right] dt, \qquad \ddot{q} = -GM \frac{q}{|q|^3}. \tag{6o}$$

The test mass cancels; the prescribed source's GM remains. For two moving gravitating bodies the relative equation contains $G(M + m)$, so the test-body qualification matters. Cancellation from these equations is not a classification as an RG-irrelevant coupling.

Under $q_\lambda(t) = \lambda q(t/\theta)$ and a correspondingly rescaled interval, free action scales by λ^2/θ at fixed m. Gravity with fixed GM requires $\theta = \lambda^{3/2}$ and its action scales by $\lambda^{1/2}$. Harmonic covariance sends w to w/θ^2. These parameter transformations are different from taking the partition mesh to zero at fixed physical w, GM and initial data.

In four-dimensional pure Yang–Mills, use the connection convention $F = dA + A \wedge A$. Give all four coordinates length units, with $x_4 = ct_E$, and write

$$\frac{S_E}{\hbar} = \frac{1}{4g^2} \int F^a_{\mu\nu} F^a_{\mu\nu} \, d^4x.$$

The classical dilation $A_\lambda(x) = \lambda^{-1} A(x/\lambda)$ gives $F_\lambda(x) = \lambda^{-2} F(x/\lambda)$; its fourth-power factor cancels the volume factor. The coupling is dimensionless, and its overall g^{-2} factor also cancels from the classical vacuum equations. Quantum running makes this normalization relevant to fluctuations. The gauge comparison already supplies $\hbar > 0$; it does not derive that assumption from Newton's trajectory.

With inverse-length renormalization scale μ, the standard weak-coupling beta function for pure $SU(3)$ is

$$\mu \frac{dg}{d\mu} = -b_0 g^3 + O(g^5), \qquad b_0 = \frac{11}{16\pi^2}.$$

The normalization is checked in Lüscher, equation (2.31) [@Luscher2010WilsonFlow] (passage read, $N = 3, N_f = 0$; §3.3 also describes quantum breaking of classical scale invariance). In the **one-loop truncated flow**, dimensional transmutation gives

$$\Lambda_1 = \mu e^{-1/(2b_0 g^2(\mu))}, \qquad g^{-2}(a_n) = g^{-2}(a_0) + 2b_0 \log(a_0/a_n), \quad \mu = 1/a. \tag{6p}$$

Differentiating g^{-2} gives $2b_0$ with respect to $\log \mu$, so Λ_1 is invariant and consecutive logarithmic increments telescope. Equation (6p) holds along every scaling sequence in the positive-coupling domain $a\Lambda_1 < 1$; every sequence $a_n \to 0$ eventually enters it. Monotonicity and fixed scaling ratios are unnecessary within that domain. Higher-loop running and scheme normalization are separate from this explicit truncated formula.

The overall factor in a retained weight. For a dimensionless fixed reference measure ν and a finite positive integral, define Euclidean or Gibbs elimination by

$$\mathcal{B}_\hbar(S)(x) = -\hbar \log \int e^{-S(x,z)/\hbar}\, d\nu(z), \qquad \hbar > 0.$$

Then the user's denominator observation remains exact under blocking:

$$e^{ic_* S/\hbar} = e^{iS/(\hbar/c_*)}, \qquad \mathcal{B}_\hbar(c_* S) = c_* \mathcal{B}_{\hbar/c_*}(S). \tag{6q}$$

The first identity concerns real-time phase; the second concerns a positive Euclidean weight. Its proof is substitution into the logarithm. Fixed normalizations independent of x change only an additive action constant. Compatible product measures make successive eliminations associate by Fubini. A reference measure that also changes with c_* must be compared separately.

Once the blocking maps and measures are specified, marginalization therefore determines the retained weight at each resolution. Reexpressing these actions at a common cutoff defines an RG transformation. It can generate interactions absent from the starting action, remain at a fixed point, or change only by normalization. Neither a differentiable flow nor closure in finitely many couplings follows from the identity. The higher-dimensional gauge interactions in §6 give its concrete use. Interpreting a positive weight as quantum theory also requires the supplied reconstruction hypotheses; the identity alone is not a derivation of quantum mechanics.

At fixed $\hbar$, multiplying S changes fluctuation weights exactly as replacing its action denominator by $\hbar/c_*$. Classical stationarity cannot distinguish those weights. In the pure-gauge connection convention, put $I[A] = \frac{1}{4}\int F^2 d^4x$ and $S_E = A_* I[A]$, with A_* in action units. Then $g^2 = \hbar/A_*$. Changing A_* to $c_* A_*$ at the same reference μ gives, within the one-loop truncated flow,

$$g^2_{c_*} = \frac{g^2}{c_*}, \qquad \Lambda_{1,c_*} = \mu e^{-c_*/(2b_0 g^2)} = \mu(\Lambda_1/\mu)^{c_*}. \tag{6r}$$

Thus an overall action factor can change a generated inverse length exponentially, even though it cancels from the classical vacuum equations. It is not a linear change of units for Λ_1.

Keeping constants explicit gives a parallel mechanical calculation. For $L, T, m, GM > 0$, write $q = Ly$, $t = Ts$ in (6o). Then

$$\frac{S}{\hbar} = \frac{mL^2}{\hbar T} \int \left[\frac{|y'|^2}{2} + \frac{GMT^2/L^3}{|y|} \right] ds.$$

The dimensionless motion parameter GMT^2/L^3 and action prefactor $mL^2/(\hbar T)$ have different roles. Choosing $T = \sqrt{L^3/(GM)}$ sets the first to one and leaves $m\sqrt{GML}/\hbar$ in the action. At the conversion length $r_g = GM/c^2$, that prefactor is $\alpha_G = GMm/(\hbar c)$. This choice introduces no relativistic dynamics into the Newtonian model. For a free body only the kinetic term and its action prefactor remain. Use $h_P = 2\pi\hbar$ for Planck's constant; the time-cell duration h in (6n) is a different quantity.

This shows consistency of running along arbitrary admitted scale sequences. It supplies no bound on the full blocked action, continuum axioms or physical spectrum. The remaining spectral requirement is

$$0 < \frac{E_{\mathrm{gap}}}{\hbar c \Lambda} < \infty,$$

with surviving finite-energy spectral weight, as in the continuum comparison. Thus the project tracks three distinct survivors: a motion or observable law independent of refinement, a physical dimensional scale, and action normalization. The harmonic curve construction proves the first in its model. Neither it nor mass cancellation selects a positive action unit, and dimensional transmutation alone proves no mass gap.

4. Gauge refinement: geometry, marginalization and reflection

Take a finite oriented lattice Γ and a refinement Γ'. For an old edge represented by a path of fine edges, define

$$p_{\Gamma'\Gamma}(U)_e = U_{e_1} \cdots U_{e_k},$$

using inverses when an orientation reverses. These maps compose exactly under further refinements and are gauge covariant: transformations at the new interior vertices cancel in the product. Holonomies of old loops are therefore represented exactly on the fine lattice.

For a probability measure the consistency equation is stronger:

$$p_{\Gamma'\Gamma *}\mu_{\Gamma'} = \mu_{\Gamma}. \tag{7}$$

At finite volume the left side can always be formed. Its effective Boltzmann weight is the integral of the fine weight over the fibres of p, with the Haar disintegration. This defines an effective action up to a normalization constant. Equation (7) with a proposed coarse action is a theorem to prove, not a consequence of having multiplied the links correctly.

Uniformly halving a D-cube produces 2^D subcubes: four, eight or sixteen for $D = 2, 3, 4$. Local subdivisions and anisotropic spacetime refinements are possible as well, provided neighbouring cells and the projection maps remain compatible. The uniform grid is a convenient sequence through a larger refinement problem.

Time refinement alone versus a field continuum. At fixed spatial spacing a_s, a Hamiltonian lattice theory has the exact temporal law $T_{a_s}(t) = e^{-tH_{a_s}/\hbar}$ and $T_{a_s}(u+v) = T_{a_s}(v)T_{a_s}(u)$. Taking the time step to zero leaves the spatial cutoff in place. Refining space adds degrees of freedom and changes the Hilbert space, so Theorem 3 on one fixed space cannot be applied without comparison maps. An isotropic Euclidean refinement removes both cutoffs together; an anisotropic route must also match electric and magnetic normalizations to recover the same physical speed c. This is the additional task in the four-cube comparison beyond inserting a time in mechanics.

One essential property can be carried exactly. Suppose a fine measure is reflection positive, p commutes with time reflection Θ, and every coarse positive-time observable pulls back to a fine positive-time observable. Then its pushforward is reflection positive:

$$\int \overline{\Theta f}\, f \, d(p_*\mu) = \int \overline{\Theta(f \circ p)}(f \circ p)\, d\mu \geq 0. \tag{8}$$

The support condition on p matters; a block straddling the reflection plane cannot be assumed to have it. Equation (8) supplies positivity for an exact blocking map with those properties. Subsequent truncation of the effective action must preserve it separately.

5. The exactly soluble subdivision law in $1+1$ dimensions

Let $G = SU(3)$ with normalized Haar measure. Choose the group metric so that $-\Delta_G \chi_R = C_2(R)\chi_R$ and $C_2(\mathbf{3}) = 4/3$. Let $\lambda_2 > 0$ have units of inverse area and define the face kernel

$$k_A(U) = \sum_R d_R \chi_R(U) e^{-\lambda_2 A C_2(R)/2}.$$

It is the heat kernel at group heat-time $\lambda_2 A$. Character orthogonality gives the exact convolution identity

$$\int_G k_{A_1}(XU^{-1}) k_{A_2}(UY)\, dU = k_{A_1+A_2}(XY). \tag{9}$$

Indeed, integrating two characters contributes $\delta_{RS}\chi_R(XY)/d_R$, so their heat exponents add and one factor d_R remains. When an interior edge splits a face, (9) integrates out precisely that new edge. Splitting an edge alone uses invariance of Haar measure under multiplication. These elementary moves yield (7) for the heat-kernel surface measure with density $Z^{-1}\prod_f k_{A_f}(U_{\partial f})$.

This is the established lattice-to-continuum construction of two-dimensional Yang–Mills. Lévy proves subdivision consistency and constructs continuum random holonomy, including the continuity needed to extend beyond a nested set of graph edges (Lévy, *Yang–Mills Measure on Compact Surfaces*, Theorem 1.6.1 and §§2.5–2.10, passage).

Here the counterpart of Newton's unequal-time insertion is division into any two positive areas, $A = A_1 + A_2$. The law already knows how to remove the new variable. It also keeps its coupling parameter: composition works for every $\lambda_2 > 0$.

The spectrum must still be identified. On a spatial circle of circumference L, a Euclidean duration t has area Lct. The physical Hilbert space is $L^2(G)^G$, the class functions, and the character expansion gives

$$E_R = \frac{\hbar c \lambda_2 L}{2} C_2(R), \qquad E_1 - E_0 = \frac{2}{3}\hbar c \lambda_2 L \quad \text{for } SU(3). \tag{10}$$

These are global electric-flux energies. Pure $1+1$-dimensional Yang–Mills has no propagating transverse gluon, and (10) grows with L. Thus the solved refinement limit supplies neither a finite infinite-line glueball mass nor the four-dimensional spectrum. Its value for this programme is the exact composition mechanism and a spectral problem whose operator and boundary conditions are explicit.

There is a dimensional mass unit $\hbar\sqrt{\lambda_2}/c$ and a string tension unit $\hbar c \lambda_2$. The absence of local particle excitations has a dynamical cause. This corrects the older dimensional claim in G07 that no mass unit can be formed in two dimensions.

Other $1+1$ gap theories sharpen the comparison. The Schwinger model, $U(1)$ gauge theory with one massless charged Dirac fermion, has an exactly soluble massive local boson (Schwinger 1962, abstract). Its axial singlet symmetry is anomalous. With several massless flavours, the spectrum instead includes a massive boson and $N_f - 1$ massless bosons (Keegan 2015, §3, passage). Coleman's theorem also constrains continuous symmetry breaking under its relativistic $1+1$ hypotheses (Coleman 1973, metadata). Thus lower dimension supplies exact theories with real spectral content, but the four-dimensional pion's symmetry-breaking mechanism must be checked afresh there. These examples serve the comparison; they do not enlarge the main proof goals.

6. What changes in $1+2$ and $1+3$ dimensions

An interior link of a hypercubic D-dimensional lattice borders $2(D-1)$ plaquettes. In $D = 2$ the two-face convolution closes. In $D = 3$ or 4, integrating a shared link couples four or six incident plaquettes. The result generally depends on several boundary loops, rather than on one coarse plaquette with one adjusted coefficient.

There is an exact local representation of this operation. Expand the face weights in characters. For chosen incident representations, the link integral is

$$P_{\mathrm{inv}} = \int_G \bigotimes_i R_i^{\sigma_i}(U)\, dU, \tag{11}$$

where a reversed orientation uses the dual representation. Haar invariance and unitarity give $P_{\mathrm{inv}}^* = P_{\mathrm{inv}}$ and $P_{\mathrm{inv}}^2 = P_{\mathrm{inv}}$, with $\|P_{\mathrm{inv}}\| \le 1$. It is the orthogonal projector onto invariant tensors. In two dimensions Schur orthogonality gives (9); in higher dimensions the additional representation labels and invariant tensors carry the boundary interaction. Formula (11) is an exact finite integration identity, not a physical transfer-matrix gap.

The analogy with (1) is now precise. Newton's quadratic elimination adds an endpoint-independent scalar; after the cubic correction, the same cell law closes. Gauge-field elimination produces an entire boundary-dependent effective interaction. A continuum argument must control that interaction and the observables along with the coupling.

Use a dimensionally explicit convention

$$\frac{S_E}{\hbar} = \frac{1}{4\lambda_D} \int F^a_{\mu\nu} F^a_{\mu\nu}\, d^D x, \qquad [\lambda_D] = \mathrm{length}^{D-4}.$$

In the classical-coupling convention of G07, $\lambda_D = \hbar g^2_{\mathrm{cl,D}}$. The dimensionless lattice coupling is $g^2_{\mathrm{lat}} = \lambda_D a^{4-D}$; in $D = 4$ it must also be run with the cutoff.

Spacetime	Refinement parameter at fixed physical coupling	Physical gap question
$1+1$	$\lambda_2 a^2$	Exact holonomy construction; circle flux spectrum (10)
$1+2$	$\lambda_3 a$	$E_{\mathrm{gap}} = C_3 \hbar c \lambda_3$, with $0 < C_3 < \infty$ to prove
$1+3$	$g_0^2(a)$, dimensionless	$E_{\mathrm{gap}} = C_4 \hbar c \Lambda$, with scale and positive finite C_4 to establish

In $1+2$ dimensions the coupling supplies an inverse length, and the ultraviolet problem is superrenormalizable. Dimensional analysis fixes the unit in the table, conditional on constructing the theory; it does not determine C_3 or its sign. At finite physical L, the form is $E(L) = \hbar c \lambda_3 f(\lambda_3 L)$, with the large-$L$ limit still to control. The small-box constant-mode approximation is a distinct limit from small lattice spacing at fixed box size.

Source boundary in three dimensions. Chandra, Chevyrev, Hairer and Shen construct a renormalized stochastic Yang–Mills–Higgs flow on the three-torus, locally in its auxiliary stochastic time, allowing possible finite-time blow-up. Global survival and an invariant measure remain open in that construction (2024 paper, introduction, passage). It therefore does not establish a finite-volume OS quantum theory or its physical gap. The earlier one-function note overstated this input; its dimensional reduction is conditional on existence.

The Karabali–Nair Hamiltonian approach supplies another mechanism, but its current-variable kinetic scale must be distinguished from a proved gap on the full physical Hilbert space (Karabali–Nair 1996, eqs. (28)–(29), passage). Nair's 2026 treatment presents an outline of a proof and identifies the gap as open (*Towards a Proof of Mass Gap in 3d Yang–Mills Theory*, abstract and introductory passage). The $1+2$ case is a genuine target with simpler scaling, not a solved theorem to import wholesale.

In $1+3$ dimensions the leading coupling is marginal. Perturbatively, one blocking step changes it by

$$g_0^{-2}(2a) = g_0^{-2}(a) - 2b_0 \log 2 + O(g_0^2(a)), \qquad b_0 = \frac{11}{16\pi^2} \quad (SU(3)).$$

The ultraviolet running sets the scale convention; the gap requires control through the regime where that expansion no longer suffices. This is the role of H1 in the existing mass-gap map, with construction and nontriviality still explicit. The dimension table supplies a hierarchy of constructive problems rather than an implication from one dimension's gap to another's.

7. What estimate would construct the gauge limit?

Here is a probability counterpart of Theorem 3. Fix a finite physical box and nested lattices Γ_n, $a_n = 2^{-n}a_0$, with compatible projections p_{mn}, $m > n$. Let μ_n be probability measures and use total variation with convention $\|\mu - \nu\|_{\mathrm{TV}} = \sup_A |\mu(A) - \nu(A)|$.

Proposition 4 (summable consistency errors). If

$$\|p_{n+1,n*}\mu_{n+1} - \mu_n\|_{\mathrm{TV}} \le e_n, \qquad \sum_{n=0}^{\infty} e_n < \infty, \tag{12}$$

then $p_{mn*}\mu_m$ converges in total variation as $m \to \infty$ to a probability measure ν_n. These limits are exactly consistent, $p_{n+1,n*}\nu_{n+1} = \nu_n$, with

$$\|\nu_n - \mu_n\|_{\mathrm{TV}} \le \sum_{j \ge n} e_j.$$

Proof. Pushforward contracts total variation, so the difference of the m-th and $(m+1)$-st projected measures is at most e_m. The series is Cauchy in the space of finite measures. Limits preserve normalization and commute with each fixed pushforward. □

The consistent measures define a measure on the inverse limit of these compact graph-configuration spaces. This is a measure on generalized holonomies for the chosen graphs. Extending it to physical continuum fields or all relevant paths, establishing Euclidean covariance and OS regularity, and proving nontriviality need additional estimates. A fixed-volume bound in (12) also needs a local version uniform in the box to support $L \to \infty$.

For $e_n \le Ca_n^\alpha$, $\alpha > 0$, the tail is bounded by $Ca_n^\alpha/(1 - 2^{-\alpha})$. The Newton estimate uses this kind of summability. In a fixed D-dimensional volume, a per-cell contribution of size $a^{D+\alpha}$ would sum to order a^α; obtaining an actual bound on normalized observables or measures from that power counting is a separate step. Marginal terms with accumulated logarithms must first be absorbed into running couplings and counterterms. A geometrically shrinking cell by itself supplies no summable estimate.

Total variation in (12) is a sufficient, potentially overstrong norm. A useful implementation can instead bound a separating class of renormalized local observables and prove tightness in a suitable topology. The topology, observables and constants must be named. In higher dimensions unsmeared loop holonomies may themselves require renormalization, so the bare graph projective limit alone is too weak a target for the physical theory.

Finally, consistency of measures supplies no energy threshold. To retain a positive physical gap, carry a uniform large-time bound

$$\langle O^* e^{-t(H-E_0)/\hbar} O\rangle_c \le K_O e^{-E_- t/\hbar}, \qquad E_- > 0,$$

for a dense family of reconstructed local vectors, with surviving finite-energy spectral weight. For the pion benchmark, the preserved Ward identity and broken-symmetry residue instead require a soft channel. The spectral criteria state the two limiting conclusions. Short-distance consistency and long-distance spectral control are complementary parts of one construction.

8. Consequence for STATE

The retained harmonic proof constructs both trajectory and force law for every admitted refinement sequence; it supplies no action normalization. Its sharper finite constants and rational

norm argument remain useful beside the broader ODE theorem. Exact Gaussian/surface composition and the summable consistency criterion answer separate law-level questions, rather than repeating that initial-value proof.

For higher-dimensional gauge fields the generated boundary interaction, response norm, cut-off/volume limits and reconstruction still need control. The SU(2) barrier and weak-strip estimates give current local results; they do not yet transport to physical recoupling and stable iteration. A full physical gap needs a further spectral estimate. Physical recording laws have their own conditional-noise, controller-memory and nonlinear-closure premises. None follows just by inserting an integration variable. New research remains paused during consolidation.